



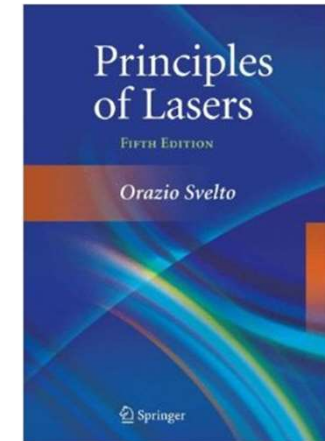
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch3-2

Resonant Interaction between Electromagnetic Field & Matter



Dr. Dan Luo(罗丹)

Email: luod@sustech.edu.cn

Office: Room 331 at College
of Engineering South Tower

Tel: 0755-88018552



Chapter 3: Resonant Interaction between Electromagnetic Field & Matter

❖ 3.2 Line broadening mechanisms

- Modification of SP, STE and STA considering the line broadening

❖ 3.3 Rate equation of typical lasers

- Rate equation for three-level system (single-mode)
- Rate equation for four-level system (single-mode)
- Rate equation for multi-mode resonance

3.2 Modification of the rate equation of SP, STA, and STE

Spectral Line function $\tilde{g}(\nu, \nu_0)$ $\Rightarrow$ Transition possibility at different frequencies

$$P(\nu) = n_2 h \nu_0 \underline{A_{21} \tilde{g}(\nu, \nu_0)} = n_2 h \nu_0 \underline{A_{21}(\nu)} \quad A_{21}(\nu) = A_{21} \tilde{g}(\nu, \nu_0)$$

$$B_{21} = \frac{c^3}{8\pi h \nu^3} A_{21} \quad A_{21}(\nu) = A_{21} \tilde{g}(\nu, \nu_0) \quad \Rightarrow \quad B_{21} = \frac{c^3}{8\pi h \nu^3} \frac{A_{21}(\nu)}{\tilde{g}(\nu, \nu_0)}$$

$$\left(\frac{dn_{21}}{dt} \right)_{sp} = A_{21} n_2 \quad \text{or} \quad B_{21} \tilde{g}(\nu, \nu_0) = B_{21}(\nu) = \frac{c^3}{8\pi h \nu^3} A_{21}(\nu)$$

$$W_{12} = B_{12} \rho_\nu \quad W_{21} = B_{21} \rho_\nu$$

$$\left(\frac{dn_{21}}{dt} \right)_{st} = W_{21} n_2$$

$$\left(\frac{dn_{21}}{dt} \right)_{sp} = \int_{-\infty}^{+\infty} n_2 A_{21}(\nu) d\nu = n_2 A_{21}$$

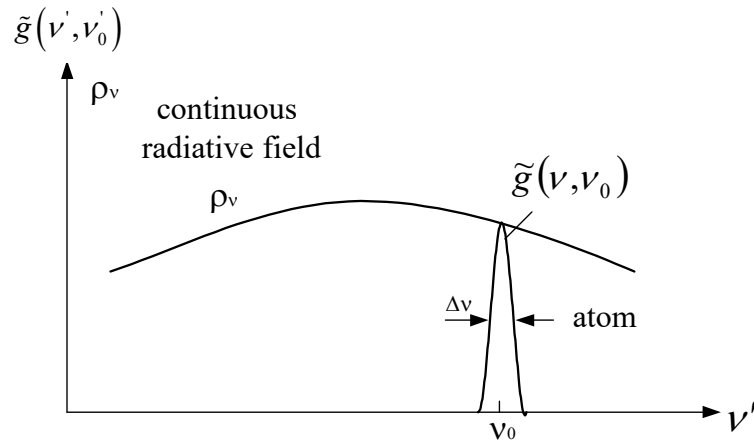
Line broadening does NOT influent SP

$$W_{21}(\nu) = B_{21}(\nu) \rho_\nu = B_{21} \tilde{g}(\nu, \nu_0) \rho_\nu$$

Modified expression $\left(\frac{dn_{21}}{dt} \right)_{st} = \int_{-\infty}^{+\infty} n_2 W_{21}(\nu) d\nu = n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho_\nu d\nu$

Two cases for $\left(\frac{dn_{21}}{dt}\right)_{st} = \int_{-\infty}^{+\infty} n_2 W_{21}(\nu) d\nu = n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho_\nu d\nu$

(1) The interaction of atoms with continuous radiation

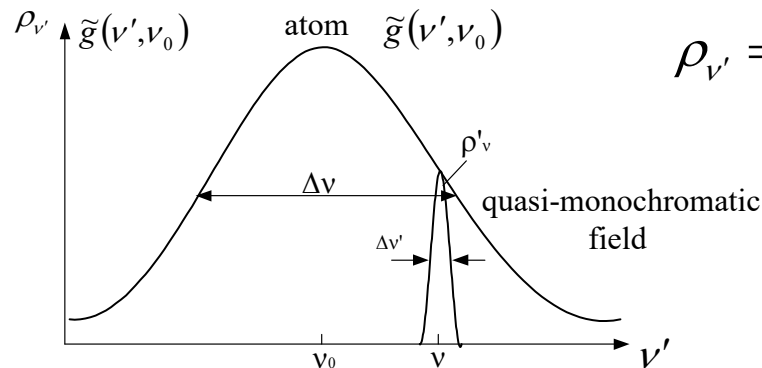


$$\Delta \nu' \gg \Delta \nu$$

$$\begin{aligned} \left(\frac{dn_{21}}{dt}\right)_{st} &= n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho_{\nu_0} d\nu \\ &= n_2 B_{21} \rho_{\nu_0} \end{aligned}$$

$$W_{21}(\nu) = B_{21} \rho_{\nu_0} \quad W_{12}(\nu) = B_{12} \rho_{\nu_0}$$

(2) The interaction of atoms with quasi-monochromatic radiation $\Delta \nu' \ll \Delta \nu$



$$\rho_{\nu'} = \rho \delta(\nu' - \nu) \quad \int_{-\infty}^{+\infty} \rho_{\nu'} d\nu' = \int_{-\infty}^{+\infty} \rho \delta(\nu' - \nu) d\nu' = \rho$$

$$\begin{aligned} \left(\frac{dn_{21}}{dt}\right)_{st} &= n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho \delta(\nu' - \nu) d\nu' \\ &= n_2 B_{21} \tilde{g}(\nu, \nu_0) \rho \end{aligned}$$

$$W_{21} = B_{21} \tilde{g}(\nu, \nu_0) \rho \quad W_{12} = B_{12} \tilde{g}(\nu, \nu_0) \rho$$

Physical Meaning: due to the line broadening, the light with frequency ν does not have to be exactly the same as the center frequency ν_0 to have stimulated recombination. Instead, the frequency in a certain range near $\nu=\nu_0$ can generate stimulated emission.

Other expression of STE and STA possibility

photon density of mode l

mode density

$$n_\nu = 8\pi\nu^2/c^3$$

$$\because \rho = N_l h\nu \quad A_{21}/B_{21} = 8\pi h\nu^3/c^3 = n_\nu h\nu = n_\nu \frac{\rho}{N_l} \Rightarrow B_{21} = \frac{N_l A_{21}}{n_\nu \rho}$$

$$W_{21} = B_{21} \tilde{g}(\nu, \nu_0) \rho = \frac{A_{21}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$W_{12} = B_{12} \tilde{g}(\nu, \nu_0) \rho = \frac{f_2}{f_1} \frac{A_{12}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{12}(\nu, \nu_0) \nu N_l$$

Emission cross-section $\sigma_{21}(\nu, \nu_0) = \frac{A_{21}\nu^2}{8\pi\nu_0^2} \tilde{g}(\nu, \nu_0)$

Absorption cross-section $\sigma_{12}(\nu, \nu_0) = \frac{f_2}{f_1} \frac{A_{12}\nu^2}{8\pi\nu_0^2} \tilde{g}(\nu, \nu_0)$

Emission cross-section reaches its maximum at central frequency

Emission cross-section of
homogenous broadening at
central frequency

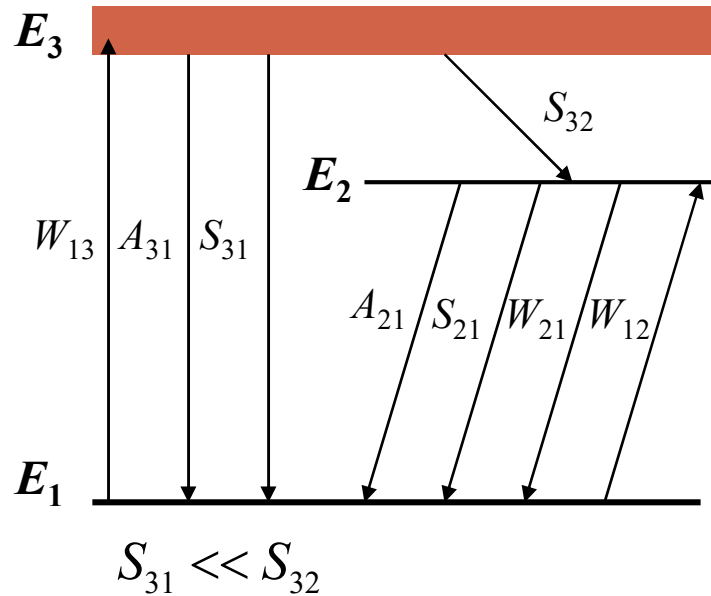
$$\sigma_{21}(\nu_0, \nu_0) = \frac{A_{21}\nu^2}{4\pi^2\nu_0^2\Delta\nu_H}$$

Emission cross-section of
inhomogeneous broadening at
central frequency

$$\sigma_{21}(\nu_0, \nu_0) = \frac{\sqrt{\ln 2} A_{21}\nu^2}{4\pi^{3/2}\nu_0^2\Delta\nu_D}$$

3.3 Rate equation of typical lasers

❖ Rate equation for three-level system (ruby laser)



$$\begin{cases} \frac{dn_3}{dt} = n_1 W_{13} - n_3 (S_{32} + A_{31} + S_{31}) \\ \frac{dn_2}{dt} = n_1 W_{12} - n_2 W_{21} - n_2 (A_{21} + S_{21}) + n_3 S_{32} \\ n_1 + n_2 + n_3 = n \end{cases}$$

Consider the cavity mode l

$$\frac{dN_l}{dt} = n_2 W_{21} - n_1 W_{12} - \frac{N_l}{\tau_{Rl}}$$

The average photon lifetime

$$I_m = I_0 e^{-2\delta m} \xrightarrow{m = \frac{t}{(2L'/c)}} I_m = I_0 e^{-\frac{2\delta c t}{2L'}} \longrightarrow I(t) = I_0 e^{-\frac{t}{\tau_R}}$$

optical length

$$\frac{1}{\tau_R} \Rightarrow \tau_R = L' / \delta c$$

$$W_{21} = B_{21} \tilde{g}(\nu, \nu_0) \rho = \frac{A_{21}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$W_{12} = B_{12} \tilde{g}(\nu, \nu_0) \rho = \frac{f_2}{f_1} \frac{A_{12}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{12}(\nu, \nu_0) \nu N_l$$

❖ Rate equation for three-level system

$$\left\{ \begin{array}{l} \frac{dn_3}{dt} = n_1 W_{13} - n_3 (S_{32} + A_{31}) \\ \frac{dn_2}{dt} = - \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - n_2 (A_{21} + S_{21}) + n_3 S_{32} \\ n_1 + n_2 + n_3 = n \\ \frac{dN_l}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \end{array} \right.$$

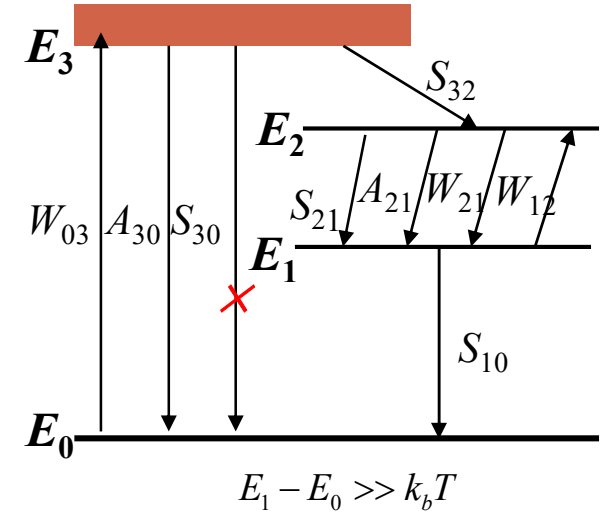
❖ Rate equation for four-level system (Nd:YAG)

$$\frac{dn_3}{dt} = n_0 W_{03} - n_3 (S_{32} + A_{30})$$

$$\begin{aligned} \frac{dn_2}{dt} &= n_3 S_{32} - n_2 (S_{21} + A_{21}) + n_1 W_{12} - n_2 W_{21} \\ &= n_3 S_{32} - n_2 (S_{21} + A_{21}) - \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l \end{aligned}$$

$$\frac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + n_3 A_{30}$$

$$\begin{aligned} \frac{dN_l}{dt} &= n_2 W_{21} - n_1 W_{12} - \frac{N_l}{\tau_{Rl}} \\ &= \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \end{aligned}$$



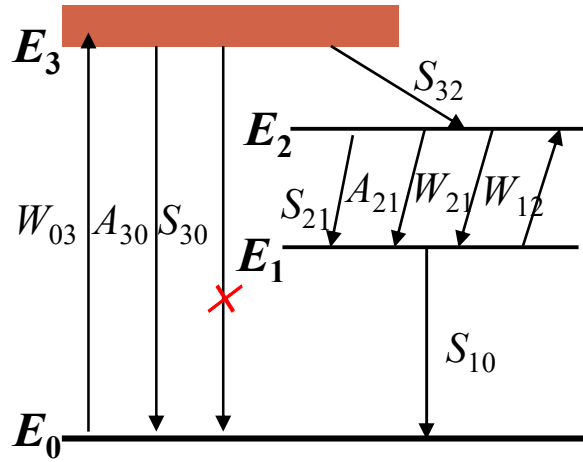
$$W_{21} = \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$n_0 + n_1 + n_2 + n_3 = n$$

❖ Rate equation for four-level system

$$\left\{ \begin{array}{l} \frac{dn_3}{dt} = n_0 W_{03} - n_3 (S_{32} + A_{30}) \\ \frac{dn_2}{dt} = n_3 S_{32} - n_2 (S_{21} + A_{21}) - \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l \\ \frac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + n_3 A_{30} \\ n_0 + n_1 + n_2 + n_3 = n \\ \frac{dN_l}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \end{array} \right.$$

- Rate equation of laser is established according to the specific physical processes
- The rate equation of a laser can have different expressions

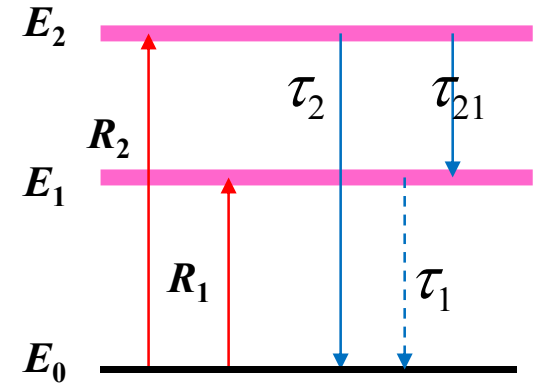


$$\frac{dn_2}{dt} = n_3 S_{32} - n_2 (S_{21} + A_{21}) + n_1 W_{12} - n_2 W_{21}$$

$$\frac{dn_1}{dt} = n_2 (S_{21} + A_{21}) - n_1 W_{12} + n_2 W_{21} - n_1 S_{10}$$

$$\frac{dn_2}{dt} = R_2 - \frac{n_2}{\tau_2} - \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$\frac{dn_1}{dt} = R_1 - \frac{n_1}{\tau_1} + \frac{n_2}{\tau_{21}} + \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l$$



R_1 and R_2 is the unit time to pump carriers from ground state to E_1 and E_2 .

τ_1 and τ_2 is the lifetime of energy level E_1 and E_2

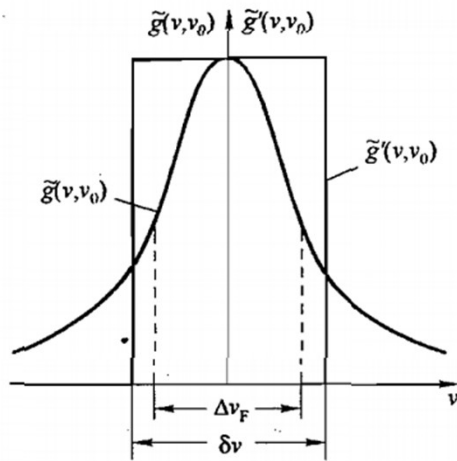
τ_{21} is the lifetime of spontaneous emission from $E_2 \rightarrow E_1$

❖ Rate equation for multi-mode resonance

$$l = l_1, l_2 \cdots l_n \quad \nu = \nu_{l_1}, \nu_{l_2} \cdots \nu_{l_n} \quad N_l = N_{l_1}, N_{l_2} \cdots N_{l_n}$$

$$\frac{dn_2}{dt} = n_3 S_{32} - n_2 (A_{21} + S_{21}) - \sum_l \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu_l, \nu_0) \nu N_l$$

$$\frac{dN_l}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \rightarrow \frac{dN_{l_1}}{dt} = \cdots, \cdots \frac{dN_{l_n}}{dt} = \cdots$$



To simplify

- No need to consider the difference between different modes
- The difference between diffraction loss can be ignored

➡ Line function treated as rectangular shape

➡ The mode loss and lifetime of modes are the same

$$\tilde{g}'(\nu, \nu_0) = \tilde{g}(\nu_0, \nu_0)$$

$$\delta \nu = 1 / \tilde{g}(\nu_0, \nu_0)$$

$$\sum_l \sigma_{21}(\nu_l, \nu_0) \nu N_l \rightarrow \sigma_{21} \nu \sum_l N_l \rightarrow \sigma_{21} \nu N$$

Simplified rate equation for multi-mode resonance

$$\frac{dn_3}{dt} = n_0 W_{03} - \frac{n_3 S_{32}}{\eta_1} \quad \eta_1 = \frac{S_{32}}{S_{32} + A_{30}} \quad E_3 \rightarrow E_2 \text{ quantum efficiency (pumping efficiency)}$$

Note: In the original image, $n_3 S_{32}$ is circled in red, and a red arrow points from this circle to the term $n_3(S_{32} + A_{30})$ in the denominator of η_1 .

$$\frac{dn_2}{dt} = -\left(n_2 - \frac{f_2}{f_1} n_1\right) \sigma_{21} \nu N - \frac{n_2 A_{21}}{\eta_2} + n_3 S_{32} \quad \eta_2 = \frac{A_{21}}{A_{21} + S_{21}} \quad E_2 \rightarrow E_1 \text{ fluoresce quantum efficiency (quantum yield)}$$

Note: In the original image, $n_2 A_{21}$ is circled in red, and a red arrow points from this circle to the term $n_2(A_{21} + S_{21})$ in the denominator of η_2 .

$$\frac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + \cancel{n_3 A_{30}}$$

$$\frac{dN}{dt} = \left(n_2 - \frac{f_2}{f_1} n_1\right) \sigma_{21} \nu N - \frac{N}{\tau_R}$$

N : total number of photons from all modes

$$n_0 + n_1 + n_2 + n_3 = n$$

$$\text{Quantum Efficiency } \eta_F = \eta_1 \eta_2 = \frac{\text{Total emitted photons}}{\text{Photon numbers absorbed by the materials through optical pumping}}$$